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Total No. of Questions : 13

B.Pharma (Sem-8)

BIOSTATISTICS AND RESEARCH METHODOLOGY

Subject Code : BP-801T

M.Code : 79764

Date of Examination : 17-11-2023

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. Answer Briefly :

   a) What are different types of Frequency Distribution?

   b) Distinguish between Correlation and Regression Analysis.

   c) What is the main difference between Simple Regression and Multiple Regression?

   d) What are 'Level of Significance' and 'Degree of Freedom' in Statistics?

   e) What are Statistical Errors?

   f) What is the need for Research?

   g) What are the properties of 'Contour Plot Graph'?

   h) What are key features of SPSS?

   i) What are advantages of a Factorial Design?

   j) Distinguish between Observational and Experimental Studies.

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SECTION-B

2. Given the bivariate distribution shown in Table, in which both X and Y are random variables.

   a) Compute the correlation coefficient;

   b) Test the sample correlation coefficient for statistical significance;

   c) If r is statistically significant, estimate the upper and lower limits of p and state your conclusion concerning the variability in X that is associated with the variability in Y.

X	55	49	30	69	50	52	50	60	41	50
Y	18	21	20	21	18	23	19	18	21	20

3. a) The data in Table are a sample of cholesterol levels taken from 12 hospital employees who were on a standard American diet and who agreed to adopt a vegetarian diet for 1 month. Serum-cholesterol measurements were made before adopting the diet and 1 month after. Compute the mean change in cholesterol.

   Serum-cholesterol levels (mg/dL) before and after adopting a vegetarian diet

Subject	Before	After
1	169	182
2	158	127
3	151	149
4	197	178
5	180	161
6	222	187
7	168	176
8	168	145
9	167	154
10	161	153
11	178	137
12	137	125

   b) Describe Hypothesis testing in Multiple Regression Models.

4. Describe various Graphical methods of Data Presentation.

SECTION-C

5. Write a detailed note on'Measures of Dispersion'.

6. Compare Binomial and Poisson's Distribution.

7. A study was conducted to test the question as to whether cigarette smoking is associated with reduced serum-testosterone levels in men aged 35 to 45. The study involved four